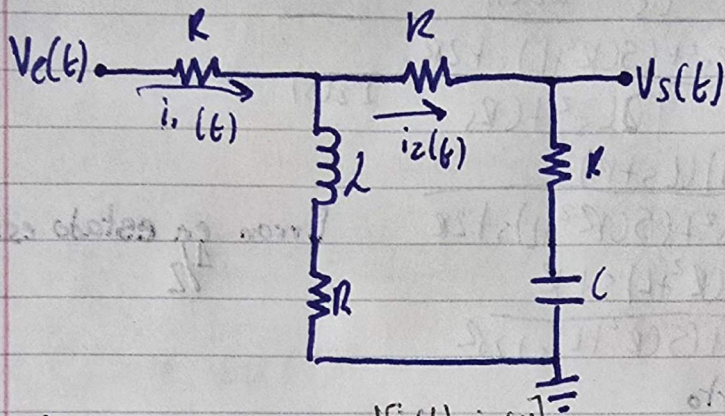


Práctica 5

15/04/2020



$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace:

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento algebraico

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L s + R} (L s + \frac{1}{C s} + 3R) I_2(s) = \frac{L s^2 + 3C R s + 1}{C L s^2 + C R s} I_2(s)$$

$$V_s(s) = (R + \frac{1}{C s}) I_2(s) = \frac{C R s + 1}{C s} I_2(s)$$

$$V_e(s) = [(L s + 2R) \frac{L s^2 + 3C R s + 1}{C L s^2 + C R s} - (L s + R)] I_2(s)$$

$$= \frac{3C L R s^2 + (3C R^2 + L) s + 2R}{C L s^2 + C R s} I_2(s)$$

Función de transferencia:

Transferencia:

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CR_s+1}{C_s} I_z(s)}{\frac{3CLR_s^2 + (SCR^2 + L)s + 2R}{(Ls^2 + CRs)} I_z(s)}$$

$$= \frac{(CR_s+1)(Ls+R)}{3CLR_s^2 + (SCR^2+1)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLRs^2 + (CR^2 + L)s + R}{3CLRs^2 + (SCR^2 + L)s + 2R}$$

F. habilidad a lazo abierto

Erron en estado estacionario
 $\frac{1}{2}$

Estabilidad en lazo abierto

$$R = 2.2 \text{ k}\Omega$$

$$L = 4.7 \text{ mH}$$

$$C = 1$$

$$3(LR)s^2 + (5(R^2 + L))s + 2R$$

$$3.102 \times 10^{-4} s^2 + 247.0047s + 4400$$

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = \frac{-(247.0047) + \sqrt{(247.0047)^2 - (4)(3.102 \times 10^{-4})(4400)}}{(2)(3.102 \times 10^{-4})}$$

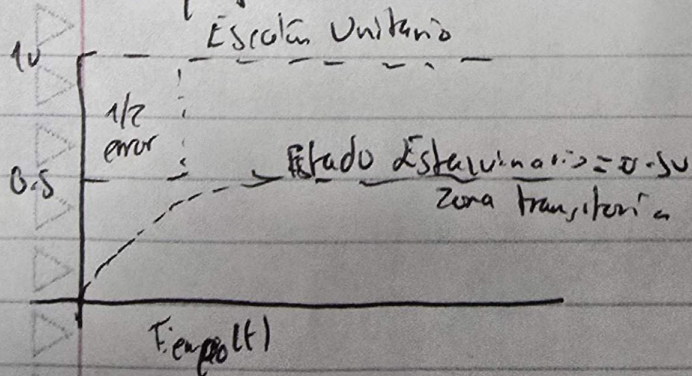
$$x_1 = 18.18185881$$

$$x_2 = \frac{-(247.0047) - \sqrt{(247.0047)^2 - (4)(3.102 \times 10^{-4})(4400)}}{(2)(3.102 \times 10^{-4})}$$

$$x_2 = -780138.8136$$

Polos Negativos Reales $\rightarrow$ Sobreamortiguado (estable)

Bosquejo:



Modelo de ecuaciones integro-diferenciales

El modelo se expresa despejando las variables dependientes:

$$V_c(t) = 2R_{i1}(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - R_{i2}(t)$$

$$i_1(t) = [V_c(t) + R_{i2}(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) = R i_2(t) + R_{i2}(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = [L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R}$$

$$i_1(t) = [V_c(t) + R i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R}$$

$$i_2(t) = [L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R}$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$\frac{V_s(s)}{V_c(s)} = \frac{(10 \mu F)(4.7 \text{ mH})(2.2 \text{ k}\Omega)s^2 + (10 \mu F)(2.2 \text{ k}\Omega)^2 + 4.7 \text{ mH}s + 2.2 \text{ k}\Omega}{(3)(10 \mu F)(4.7 \text{ mH})(2.2 \text{ k}\Omega)s^2 + [(5)(10 \mu F)(2.2 \text{ k}\Omega)^2 + 4.7 \text{ mH}]s + 2(2.2 \text{ k}\Omega)}$$

$$= \frac{1.034 \times 10^{-4} s^2 + 48.4047 s + 2200}{3.102 \times 10^{-4} s^2 + 247.0047 s + 4400}$$